THE ERDŐS SUMS OF ALMOST PRIMES: THE RÔLE OF THE PRIME 2

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Abstract. For $k \geq 1$ let $f_k = \sum_{\Omega(n)=k} 1/(n \log n)$, where Ω counts prime factors with multiplicity. Banks and Martin conjectured that $f_1 > f_2 > f_3 > \cdots$. We prove that the sequence is in fact strictly *increasing* from $k = 6$ onwards, and that the sums restricted to odd n are strictly *decreasing* for every $k \geq 1$. Taken together these say that the failure of the Banks–Martin conjecture is caused entirely by the prime 2. Both results are obtained from a single mechanism: writing the generating Dirichlet series as $\zeta(s)^z H(s, z)$, the behaviour of f_k is governed by whether the evaluation point $z = 2$ lies inside or on the boundary of the disc of analyticity of $H(s, \cdot)$. The analytic reductions are formalised in Lean 4 and the numerical inputs are certified in ball arithmetic.

1. Introduction

For $k \geq 1$ let

$$f_k \;=\; \sum_{\Omega(n)=k} \frac{1}{n \log n},$$

where $\Omega(n)$ counts the prime factors of n with multiplicity. The weight $1/(n \log n)$ is Erdős's [3], who proved that $\sum_{n \in A} 1/(n \log n)$ is bounded uniformly over all primitive sets A; each f_k converges, and $f_k \to 1$. Writing $\mathbb{N}_k = \{n : \Omega(n) = k\}$, so that $\mathbb{N}_1$ is the set of primes, Zhang [5] proved $f_k < f_1$ for every $k > 1$, and Bayless, Kinlaw and Klyve [6] showed $f_2 > f_3$. Banks and Martin [1] predicted that the pattern continues,

$$f_1 > f_2 > f_3 > \cdots, \tag{1}$$

as part of a considerably more general conjecture.

Gorodetsky, Lichtman and Wong [2] showed that the conjecture is false, by establishing the asymptotic

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + O(k \log(k+1))\big), \qquad a_1 = \tfrac{1}{4} d \log 2 > 0,$$

from which $f_k < f_{k+1}$ for all sufficiently large k. The asymptotic does not by itself locate the crossover: the implied constant in the error term is not explicit, so the argument gives no k beyond which monotonicity is guaranteed, and in particular does not decide the small cases where the conjecture might still have held.

Our first result closes that gap completely.

The proof has two halves. For k beyond an explicit threshold we bound each source of error in the asymptotic separately and show the resulting criterion is positive and increasing; below that threshold the inequality is verified by certified computation, in three overlapping ranges whose union with the analytic range is all of $[6, \infty)$.

Our second result concerns the same sums with the prime 2 removed. Write f_k^{odd} for the sum of $1/(n \log n)$ over odd n with $\Omega(n) = k$.

Together the two theorems isolate the rôle of the prime 2: the unrestricted sums eventually increase, the odd sums decrease throughout, and the difference is caused by 2 alone. This is

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not a coincidence of two separate calculations. Both follow from a single mechanism, which we isolate in §3: writing the generating Dirichlet series as $\zeta(s)^z H(s,z)$, the disc of analyticity of $H(s,\cdot)$ has radius q^s where q is the least prime involved, while the evaluation point relevant to the 2^{-k} scale is $z=2$. For the unrestricted sums $q=2$ and the evaluation point sits *on* the boundary at $s=1$, producing a factor k^2 that competes with 2^{-k}; for the odd sums $q=3$ and it lies strictly inside, so the leading behaviour is a constant times 2^{-k} with nothing competing. The k^2 factor is precisely what makes the first theorem hard and its absence is precisely what makes the second easy.

This suggests a general statement, recorded as Conjecture 8.1 in §8: for sums restricted to n with all prime factors in a set $\mathcal{P}$, the eventual increase should occur exactly when $2 \in \mathcal{P}$.

The analytic reductions in both theorems are formalised in Lean 4, and the numerical inputs are computed in rigorous ball arithmetic. We are explicit in §9 about which is which: the formalisation takes the certified constants and sweeps as hypotheses and proves the reduction from them, so that what is machine-checked and what is computed are cleanly separated. Two facts needed along the way are absent from Mathlib and may be of independent use: that $\zeta(s) - 1/(s-1) \to \gamma$ as $s \to 1^+$, and that $\zeta(s) - 1/(s-1)$ extends analytically across $s=1$.

Theorem 1.1. *For every $k \geq 6$ we have $f_k < f_{k+1}$.*

Theorem 1.2. *Let $f_k^{\mathrm{odd}} = \sum_{n \text{ odd},\ \Omega(n)=k} 1/(n \log n)$. Then $f_k^{\mathrm{odd}} > f_{k+1}^{\mathrm{odd}}$ for every $k \geq 1$, and $f_k^{\mathrm{odd}} \to 1/2$.*

2. Notation and the integral representation

Throughout, p denotes a prime and $\Omega(n)$ the number of prime factors of n counted with multiplicity. For $s > 1$ set

$$P_k(s) = \sum_{\Omega(n)=k} n^{-s}, \qquad P_k^{\mathrm{odd}}(s) = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} n^{-s}.$$

Lemma 2.1. *For $k \geq 1$,*

$$f_k = \int_1^\infty P_k(s)\, ds, \qquad f_k^{\mathrm{odd}} = \int_1^\infty P_k^{\mathrm{odd}}(s)\, ds. \tag{2}$$

Proof. For $n \geq 2$ we have $\int_1^\infty n^{-s}\, ds = 1/(n \log n)$, since $n^{-s} = e^{-s \log n}$ and $\log n > 0$. Both index sets consist of integers $n \geq 2$ (indeed $n \geq 2^k$, since $\Omega(n) = k$), and all terms are non-negative, so Tonelli's theorem permits the interchange of sum and integral. This is Lemma 2.1 of [2] in the unrestricted case; the argument is insensitive to which set of n is summed over. □

The generating functions are Euler products,

$$\sum_{k \geq 0} P_k(s) z^k = \prod_p (1 - zp^{-s})^{-1}, \qquad \sum_{k \geq 0} P_k^{\mathrm{odd}}(s) z^k = \prod_{p \geq 3} (1 - zp^{-s})^{-1}. \tag{3}$$

3. The mechanism

Both products in (3) are written in the form $\zeta(s)^z H(s,z)$. For the unrestricted product,

$$H(s,z) = \prod_p (1 - zp^{-s})^{-1} (1 - p^{-s})^z, \tag{4}$$

and for the odd product,

$$H^{\mathrm{odd}}(s,z) = (1 - 2^{-s})^z \prod_{p \geq 3} (1 - zp^{-s})^{-1} (1 - p^{-s})^z. \tag{5}$$

Proposition 3.1. *$H(s,\cdot)$ is analytic in $|z|<2^s$, and $H^{\mathrm{odd}}(s,\cdot)$ is analytic in $|z|<3^s$.*

Proof. The p-th factor of either product is $(1-zp^{-s})^{-1}(1-p^{-s})^{z}$, whose only singularity in z is the pole at $z=p^s$. The radius of the largest disc free of poles is therefore q^s, where q is the least prime occurring in the product: $q=2$ in (4) and $q=3$ in (5).

That the product converges on that disc is the point of the regularisation. Writing $x=p^{-s}$, the logarithm of the p-th factor is

$$-\log(1-zx)+z\log(1-x)=(z-z)x+O\big((|z|^2+|z|)x^2\big),$$

the first-order terms cancelling identically. Hence each factor is $1+O(p^{-2s})$ uniformly for z in a compact subset of $|z|<q^s$, and $\sum_p p^{-2s}$ converges for $s>1/2$. The product is therefore analytic there, being a locally uniform limit of analytic functions. □

Remark 3.2. The cancellation of the first-order terms is what distinguishes H from the bare Euler product: the latter has no disc of convergence containing $z=2$ at $s=1$ in either case.

This is the whole difference between the two theorems. At $s=1$ the evaluation point relevant to the 2^{-k} scale is $z=2$, and

- for the unrestricted sums $z=2$ lies *on* the boundary $|z|=2^s$ at $s=1$;
- for the odd sums $z=2$ lies strictly *inside* $|z|<3^s$.

On the boundary one picks up a factor k^2 competing with 2^{-k}; this is the source of the a_1k^2 term in (13) below, of the turnaround at $k=6$, and of the error-accounting machinery of §§5–7. Strictly inside, the leading behaviour is a constant times 2^{-k} with nothing competing, and the proof is correspondingly short (§4).

Remark 3.3. Nothing above is special to the prime 2 as such: what matters is that 2 is the smallest prime, so that removing it moves the radius from 2^s to 3^s past the fixed evaluation point $z=2$. See Conjecture 8.1.

4. The odd sums

Throughout this section $H=H^{\mathrm{odd}}$ as in (5).

4.1. **The expansion.** Substituting $u=\log\zeta(s)$ in (2) and expanding

$$s-1=e^{-u}+\gamma e^{-2u}+O(e^{-3u}), \tag{6}$$

the contributions to f_k^{odd} organise by powers of e^{-u}. Using $\int_0^\infty u^k e^{-ju}\,du/k!=j^{-(k+1)}$:

$$\text{order } e^{-u}: \quad H(1,1), \tag{7}$$

$$\text{order } e^{-2u}: \quad 2^{-(k+1)}\big[\,\partial_s H(1,2)+2\gamma H(1,2)\,\big], \tag{8}$$

$$\text{order } e^{-ju}: \quad O(j^{-(k+1)}), \quad j\ge 3. \tag{9}$$

Remark 4.1. The substitution is $u=\log\zeta(s)$, not $s=1+e^{-u}$: only the former makes the powers of u appearing in the expansion of P_k^{odd} exact powers of the integration variable, so that the Gamma integrals apply termwise. The Jacobian is $ds/du=\zeta/(-\zeta')$, which by (6) equals $e^{-u}+2\gamma e^{-2u}+O(e^{-3u})$; the factor 2γ is one γ from the Jacobian's own expansion in ε and one from (6).

4.2. **The limit.**

Lemma 4.2. *$H(1,1) = 1/2$, and hence $f_k^{\mathrm{odd}} \to 1/2$.*

Proof. Put $z = s = 1$ in (5). The general factor becomes $(1-p^{-1})^{-1}(1-p^{-1}) = 1$, so the product over $p \geq 3$ is identically 1, and $H^{\mathrm{odd}}(1,1) = (1-2^{-1})^1 = 1/2$. The claim $f_k^{\mathrm{odd}} \to 1/2$ is then (7), the orders e^{-ju} with $j \geq 2$ contributing $O(2^{-k})$ by (8) and (9). □

Remark 4.3. The same computation for the unrestricted product gives $H(1,1) = 1$, recovering $f_k \to 1$.

4.3. **The leading term.**

Lemma 4.4. *$H(1,2) = \frac{1}{4}\prod_{p\geq 3}(1-1/p)^2(1-2/p)^{-1} = 1/(4C_2) = d$, the constant of* [2]*, and*

$$\partial_s \log H(1,2) = 2\log 2 - 2S, \qquad S = \sum_{p\geq 3} \frac{\log p}{(p-1)(p-2)}.$$

Proof. Setting $z = 2$, $s = 1$ in (5) gives the stated product directly, the prefactor being $(1-2^{-1})^2 = 1/4$; that this equals $1/(4C_2)$, with C_2 the twin-prime constant, is the definition of d in [2].

For the derivative, take logarithms in (5) at $z = 2$:

$$\log H^{\mathrm{odd}}(s,2) = 2\log(1-2^{-s}) + \sum_{p\geq 3}\Big[-\log(1-2p^{-s}) + 2\log(1-p^{-s})\Big].$$

By the estimate in the proof of Proposition 3.1 each summand is $O(p^{-2s})$, and differentiating term by term is justified for $s > 3/4$ by local uniform convergence of the differentiated series, whose p-th term is $O(\log p \cdot p^{-2s})$.

Differentiating the prefactor, $\partial_s 2\log(1-2^{-s}) = 2\,\frac{2^{-s}\log 2}{1-2^{-s}}$, which at $s = 1$ is $2\log 2$. Differentiating the p-th summand,

$$\frac{-2\log p \cdot p^{-s}}{1-2p^{-s}} + \frac{2\log p \cdot p^{-s}}{1-p^{-s}} \xrightarrow{s=1} \frac{-2\log p}{p-2} + \frac{2\log p}{p-1} = \frac{-2\log p}{(p-1)(p-2)}.$$

Summing gives $\partial_s \log H^{\mathrm{odd}}(1,2) = 2\log 2 - 2S$. □

Remark 4.5. Note that $s = 1$ is interior to the region of convergence here, unlike for ζ: the product over $p \geq 3$ converges for $s > 3/4$, since $p^s > 2$ for $p \geq 3$ and $s > 3/4$ (as $3^{3/4} > 2$), so no factor degenerates.

Substituting Lemma 4.4 into (8):

$$f_k^{\mathrm{odd}} = \tfrac{1}{2} + \beta\, 2^{-k} + R_k, \qquad \beta = d\,(\log 2 - S + \gamma). \tag{10}$$

Numerically $\beta = 0.15563$. As a check independent of the derivation, Richardson extrapolation of $B(k) = 2^k(f_k^{\mathrm{odd}} - 1/2)$ with ratio $3/4$ on the certified values gives 0.157522, 0.156515, 0.156231 at $k = 16, 18, 19$, converging to β from above.

4.4. **The remainder.** Factor out the pole nearest the origin:

$$\widetilde{H}(s,z) = H(s,z)\,(1 - z\,3^{-s}),$$

analytic in $|z| < 5^s$. Cauchy's estimate on $|z| = R < 5$ gives $|[z^k]\widetilde{H}(1,z)| \leq M_R R^{-k}$ with $M_R = \sup_{|z|=R,\ s\in[1,2]} |\widetilde{H}(s,z)|$, and hence, summing (9) over $j \geq 3$,

$$|R_k| \leq M_R R^{-k}. \tag{11}$$

4.5. **Choice of radius.** R may be taken anywhere below 5. A larger R lowers the threshold K_0; a smaller R lets the Euler tail start at a smaller prime, which matters for the formalisation.

R	M_R (certified)	K_0	tail starts at
4.0	374.367	—	$p \geq 7$
4.5	826.705	—	$p \geq 11$

The tail constraint is that the p-th factor's correction be bounded by 1, which at radius R and prime P reads $R^2/(1-R/P)+1.25R \leq P^2$. At $R=4.5$ this fails for $P=7$ (giving $62.3>49$) and first holds at $P=11$.

Neither radius is used below. The Cauchy estimate is a valid upper bound but not the rate the remainder exhibits: against the certified values of §A the bound $M_R(2/9)^k$ with $M_R = 826.705058$ fails from $k=13$ onward, short by a factor of 60 at $k=20$. The derivation behind it is also unsound: the order expansion of §4.1 is asymptotic, not convergent — the order-m term is $O(k^{m-1}3^{-k})$ — so it cannot be summed over m. Truncating with a Lagrange remainder instead, and bounding the resulting coefficient sum by $\binom{k+3}{3}S$ with $S=\sum_i |g_i|3^i$ over the coefficients of $G_2 = \partial_s^2 H\,(1-z/3)^3$, gives

$$|R_k| \;\leq\; C\,(k+3)^3 3^{-k}, \qquad C = 4900. \tag{12}$$

This is derived, not calibrated. Three contributions make up C, and they are of very different sizes: the first multiplies the leading term by $C_E/3 = 1.83$ and is what sets the threshold; the second is smaller than the main term by five to eight orders of magnitude over the range that matters; the third is a cost of the splitting performed here rather than a feature of the underlying estimate.

The Lagrange remainder. Bounding $|h_j(s) - h_j(1) - (s-1)b_j|$ by $\frac{1}{2}\sup|h_j''|(s-1)^2$ and integrating $\int_0^\infty u^m e^{-3u}/m!\,du = 3^{-(m+1)}$ gives $\frac{1}{3}3^{-k}\sum_{j\leq k}|h_j''|3^j$, and $\sum_{j\leq k}|h_j''|3^j \leq \binom{k+3}{3}S$ with $S=\sum_i|g_i|3^i \leq 14414$ over the coefficients of $G_2 = \partial_s^2 H\,(1-z/3)^3$. The splitting of the integrand leaves not one error term but three: besides $R_j J$, also $h_j(1)(J - e^{-u} - 2\gamma e^{-2u})$ and $b_j(\varepsilon J - e^{-2u})$. Writing C_E for the combined constant of the three, $C_E \leq 5.48910$ over the range $s \in (1,\, s(3)]$ that the splitting requires, and this part of C is $C_E S/18 = 4395.5$.

The two coefficient tails. The main terms integrate to the partial sums over $j \leq k$, so $\sum_{j>k} h_j(1)$ and $2^{-(k+1)}\sum_{j>k}(b_j + 2\gamma h_j(1))2^j$ join R_k. The same pole-cancellation device at first and second order gives $S_0 = \sum_i |a_i|3^i \leq 44.5987$ with $a_i = [z^i]H(1,z)(1-z/3)$ and $S_1 \leq 486.1945$ with $(1-z/3)^2$ against $\partial_s H$, contributing $S_0/2 = 22.3$ and $9S_1/128 = 34.2$.

The near region. The expansions used above are asymptotic, valid for $u \geq U$; the splitting therefore needs the complementary range handled separately, which the estimate in its unsplit form does not. With $U=3$ the range $0 < u \leq U$ contributes $C_N \leq 367.8435$ from the coefficient sum and $N_U \leq 26.0962$ from the two main terms.

Together $C_E S/18 + S_0/2 + 9S_1/128 + C_N + N_U = 4846.0$, and we state (12) with $C = 4900$. All five constants are certified interval computations; see §A.

4.6. **Assembly.** From (10),

$$f_k^{\mathrm{odd}} - f_{k+1}^{\mathrm{odd}} = \beta\, 2^{-(k+1)} + (R_k - R_{k+1}),$$

so it suffices that $|R_k| + |R_{k+1}| < \beta 2^{-(k+1)}$, for which $|R_k| < \beta 2^{-k}/3$ is enough. With (12):

k	$\beta 2^{-k}/3$	$C(k+3)^3 3^{-k}$	
57	3.600×10^{-19}	6.328×10^{-19}	fails
58	1.800×10^{-19}	2.217×10^{-19}	fails
59	8.999×10^{-20}	7.759×10^{-20}	holds
60	4.500×10^{-20}	2.713×10^{-20}	holds

The ratio $T/P = C(k+3)^3(2/3)^k$ is antitone (the step ratio is $((k+4)/(k+3))^3(2/3) < 1$ for $k \geq 4$), equal to 0.0476 at $k = 59$, so the inequality holds for all $k \geq 59$. For $k = 1, \ldots, 58$ the differences are positive by direct certified computation, which reaches $k = 60$. The ranges overlap at $59, 60$, so Theorem 1.2 holds for every $k \geq 1$. □

5. The unrestricted sums: the criterion

We take as given the asymptotic of [2]:

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + O(k \log(k+1))\big), \qquad a_1 = \tfrac{1}{4} d \log 2. \tag{13}$$

Write $f_k = 1 - 2^{-k}(a_1 k^2 + E(k))$, so that E collects the entire secondary error. Then

$$f_{k+1} - f_k = 2^{-(k+1)}\big[\, a_1(k^2 - 2k - 1) + 2E(k) - E(k+1) \,\big], \tag{14}$$

so $f_{k+1} > f_k$ exactly when

$$a_1(k^2 - 2k - 1) + 2E(k) - E(k+1) > 0. \tag{15}$$

Explicitly, $E(k) = 2^k(1 - f_k) - a_1 k^2$, and (15) is an identity rather than an estimate: it holds for any choice of a_1 whatever, the content being entirely in the lower bound for $2E(k) - E(k+1)$ obtained in §6.

5.1. The source bound.

Write

$$\begin{aligned} R(k) \;&= \tfrac{\log 2}{4}\big[(2k-2)U_{\max} + 2V_{\max} + 4U_{\max}\big] \\ &\quad + S^*\big[2D(k) + D(k+1)\big] + \alpha_+(3k+4) + 3\Big(\tfrac{1.45}{k} + 10^{-3}\Big), \end{aligned} \tag{16}$$

where $D(\cdot)$ is the δ-table bound of §6, $U_{\max}, V_{\max}$ bound the signed sums

$$U_k = \sum_{i \leq k} i\, 2^i \operatorname{Re} c_i, \qquad V_k = \sum_{i \leq k} i^2\, 2^i \operatorname{Re} c_i, \qquad c_i := [z^i] G_2(1, z),$$

$S^* = \sum_i 2^i |c_i|$, and α_+ is the s-uniform bound on the first s-derivatives of the coefficients. The four groups of terms in (16) correspond to the four sources: the signed U, V combination, the δ-table convolution, the S_1 term, and the exponentially small remainder $E_{\exp}$.

Proposition 5.1. *If $|2E(k) - E(k+1)| \leq R(k)$, then the left-hand side of (15) is at least*

$$\varphi(k) \;:=\; a_1(k^2 - 2k - 1) - R(k).$$

In particular $\varphi(k) > 0$ implies $f_k < f_{k+1}$.

Proof. Immediate from (15) and the triangle inequality. □

6. Bounding the sources

We follow [2] §3 throughout. Writing $F(s,z) = (s-1)^{-z}G(s,z)$ and $G_2(s,z) = (1 - z2^{-s})G(s,z)$, their (3.6) and (3.8) give

$$I'_k = 2I'_{k,2} - \tfrac{\log 2}{4}2^{-k}B, \qquad B = \sum_{i=0}^{k} \frac{2^i}{i!}\Big[(k-i)^2 + O\big((k-i+1)\log(k-i+2)\big)\Big]G_2^{(0,i)}(1,0),$$

with $I'_{k,2} = 1/2 + O(3^{-k})$, and separately $f_k = I_k + O(1/(k2^k))$ with $I_k - I'_k = O(k/2^k)$. Tracking the powers of 2^k through these substitutions gives, exactly,

$$E(k) = E_{S_1}(k) + \tfrac{\log 2}{4}(V - 2kU) + \tfrac{\log 2}{4}\Delta(k) + E_{\exp}(k), \tag{17}$$

where $c_i := [z^i]G_2(1,z) = G_2^{(0,i)}(1,0)/i!$ and the four terms are treated in §§6.1–6.4 below.

6.1. **The S_1 term.** Set $E_{S_1}(k) := 2^k(I_k - I'_k)$. [2] bound $|G_2^{(0,i)}(s,0) - G_2^{(0,i)}(1,0)| \le (s-1)b_i$ for $s \in [1,2]$, where $b_i := \max_{s\in[1,2]} |G_2^{(1,i)}(s,0)|$, giving

$$2^k(I_k - I'_k) \;\le\; \sum_{i\ge 1} \big|[z^i]G_2^{(1,0)}(1,z)\big|\,(k+1-i)\,2^{i-1}.$$

The quantity α_+ of §5 is a bound, uniform over $s \in [1,2]$, for the coefficients $[z^i]G_2^{(1,0)}(s,z)$ — not merely their values at $s = 1$. This distinction is not cosmetic: the maximum over $s \in [1,2]$ drifts slightly into the interior for larger i, so the endpoint values do not suffice. The uniform bound is obtained from the identity

$$\log G_2(s,z) = z\big[\log(1-2^{-s}) + S_1(s) + P(s,1)\big] + \sum_{j\ge 2} \frac{z^j P(s,j)}{j},$$

with $P(s,j) := \sum_{p>2} p^{-js}$ and $S_1(s) := \sum_{p>2}\log(1-p^{-s})$, evaluated with s itself carried as a certified interval. Note that the linear-in-z term receives contributions from *both* Euler factors.

The evaluation partitions $s \in [1,2]$ into narrow sub-intervals and runs the coefficient recursion directly with s carried as an interval, rather than using Cauchy contour estimates: the latter are dominated by proximity to the pole at $s = 1$ and lose the cancellation that makes the coefficients small, by roughly two orders of magnitude.

Writing $\beta_i := \max_{s\in[1,2]} |[z^i]G_2^{(1,0)}(s,z)|$ and

$$\alpha_+ \;\ge\; \tfrac{1}{2}\sum_{i\le m} \beta_i 2^i \qquad \text{for every } m, \tag{18}$$

the estimate of this subsection is

$$|E_1(k)| \;\le\; \alpha_+(k+1), \tag{19}$$

which follows from the displayed sum by $(k-i+1) \le (k+1)$ for every i in range. The certified sweep gives $\alpha_+ \le 15.4257$.

The sweep is run with primes to $P = 10^5$, 1600 boxes in s, at 300 bits of precision, giving $\alpha_+ = 15.425661$. Two features of the computation are worth recording. First, the maximum over $s \in [1,2]$ is not always attained at $s = 1$: it drifts into the interior for larger i, so the endpoint values do not suffice and the s-uniformity had to be closed by direct evaluation rather than assumed. Second, the recursion must be run with s as an interval rather than replaced by a contour estimate, for the reason given above.

The certified coefficients decay geometrically:

i	β_i	$\widehat{\gamma}_i$
1	1.67926	2.79716
2	1.70018	6.38236
3	0.93613	5.41714
4	0.37329	2.90307
5	0.11407	1.12437
6	3.0352×10^{-2}	0.35601
7	6.6304×10^{-3}	9.2248×10^{-2}
8	1.4954×10^{-3}	2.2660×10^{-2}
9	2.6766×10^{-4}	4.7018×10^{-3}
10	7.5006×10^{-5}	1.2463×10^{-3}
12	6.3579×10^{-6}	1.0829×10^{-4}
15	2.6304×10^{-7}	5.3099×10^{-6}

Both sequences peak at small i — β_i at $i = 2$, $\widehat{\gamma}_i$ at $i = 2$ as well — and thereafter fall off by roughly a factor 4 per step, so the weighted sums $\sum_i \beta_i 2^i$ and $\sum_i \widehat{\gamma}_i 2^i$ converge comfortably and the tails beyond the computed head are geometric.

6.2. **The S_3 term: signed, not bounded.** The combination $V - 2kU$ arises from the exact algebraic identity $(k-i)^2 - k^2 = i^2 - 2ik$, so that $\sum_i 2^i c_i(i^2 - 2ik) = V - 2kU$; it is computed exactly from the certified coefficients rather than bounded. This matters because the c_i alternate in sign and the sums are correspondingly small: $U = 0 \pm 0.137$ and $V = 0 \pm 2.72$, against $\sum_i |c_i| 2^i \approx 6.98$. The identity check $\sum_i c_i 2^i = G_2(1, 2) = d$ confirms the certified head-plus-tail sum.

Combining the two indices,

$$2(V - 2kU) - \big(V - 2(k+1)U\big) = V + 2U - 2kU,$$

whose worst case over the certified balls is $\geq -V_{\max} + (2 - 2k)U_{\max}$: the coefficient of U is $2 - 2k < 0$ for $k \geq 2$, forcing the adverse choice $U = +U_{\max}$, while V's coefficient is $+1$, forcing $V = -V_{\max}$. This is an exact optimisation over the interval box, not a triangle inequality. The contribution is scaled by $\log 2/4$, as is $\Delta(k)$ below, both arising through the same $-\frac{\log 2}{4} 2^{-k} B$ route.

6.3. **The S_2 term: evaluating the Lemma 3.3 error.** Let $\delta_m := A_m - 2^{m+1} + \frac{\log 2}{4} m^2$ denote the error term of [2]'s Lemma 3.3, and $\Delta(k) := \sum_i 2^i c_i \delta_{k-i}$ its convolution against the coefficients — the error is incurred at every index $k - i$ appearing in B, not once at k.

The point of this subsection is that δ_m can be *evaluated*, not merely bounded. Setting $B_j(u) := 2^{-je^{-u}} - 1 + j \log 2\, e^{-u} \geq 0$, one has the exact decomposition

$$\delta_m = -1 - \tfrac{\log 2}{4} m + \sum_{i=0}^{m} 2^i \beta_i(m - i), \qquad \beta_i(j) := \frac{1}{i!} \int_0^\infty u^i e^{-u} B_j(u)\, du \geq 0. \tag{20}$$

Since every $\beta_i(j) \geq 0$ and B_j is decreasing in u, endpoint bracketing gives certified enclosures of δ_m, with exact incomplete-gamma panel masses for the u-integration.

Bounding $\Delta(k)$ instead by $S^* \max_m |\delta_m|$, using [2]'s stated $|\delta_m| \leq Cm \log(m + 1)$, discards the same sign cancellation exploited in §6.2 and costs roughly two orders of magnitude.

In the formalisation the bound actually used is the uniform one,

$$D(m) := (\log 2)\, m \log(m + 1) + 1, \qquad |\delta_m| \leq D(m), \tag{21}$$

and (21) is *proved* rather than assumed: the upper bound follows because the $\beta_i(j)$ in (20) are non-negative and $c^*(1) < 0$, the lower because $(\log 2)m\log(m+1) \geq \frac{\log 2}{4}m$ for $m \geq 1$, and $m = 0$ anchors at $\delta_0 = -1$, $D(0) = 1$. Thus $S^*[2D(k)+D(k+1)]$ in (16) is the unsigned route: it discards the cancellation described above, at the cost of a larger threshold, in exchange for a bound that requires no δ-table at all.

Remark 6.1. The signed evaluation of (20) is what the certified δ-table computes, and it is what brings the threshold down; the formalised route uses (21) instead. The two are alternatives, not a chain: (21) is weaker but self-contained.

6.4. **The exponentially small term.** $E_{\exp}(k)$ collects three contributions: the $s \geq 2$ tail $2^k \cdot O(1/(k2^k))$, the $2^k \cdot O(3^{-k})$ piece from $I'_{k,2} = 1/2 + O(3^{-k})$, and the $\frac{\log 2}{4}k^2 \cdot O((3/2)^{-k})$ correction from $B' = d + O((3/2)^{-k})$. Each is bounded by transferring $\sup|H| \leq 101.245$, certified on $|z| = 4.2$, to the relevant inner circles by the maximum principle. All are below 10^{-180} in the range of interest, so that the constant $3(1.45/k + 10^{-3})$ appearing in (16) is enormously conservative.

6.5. **The certified constants.** From `NumericalFacts`:

constant	certified bound	rôle
d	$\lvert 0.3786950 - d\rvert \leq 5.3 \times 10^{-8}$	$a_1 = \frac{1}{4}d\log 2$
ρ	$\lvert -0.1749323 - \rho\rvert \leq 6.6 \times 10^{-8}$	tail bounds
S^*	≤ 6.9763	§6.2
α_+	15.425661	β_i sweep, §6.1
Γ	104.5196	$\widehat{\gamma}_i$ sweep, Theorem 6.1
$U_{\max}$	0.08436	§6.2, valid $k \geq 900$
$V_{\max}$	1.907462	§6.2, valid $k \geq 900$
K_1	2433	analytic threshold

Here α_+ is as in §6.1. The constant Γ governs the refined form of the S_1 estimate: writing $\widehat{\gamma}_i := \max_{s\in[1,2]} |[z^i]G_2^{(2,0)}(s,z)|$ for the s-uniform bound on the *second* s-derivatives, set

$$\Gamma := \tfrac{1}{2}\sum_{i\geq 1} \widehat{\gamma}_i\, 2^i.$$

Applying Taylor's theorem with second-order remainder to g_i on $(1,2)$ — that is, $g_i(s) = c_i + (s-1)\widehat{b}_i + R_i(s)$ with $|R_i(s)| \leq (s-1)^2\widehat{\gamma}_i/2$ — replaces the first-order estimate of §6.1 by one whose remainder is quadratic in $s-1$.

What makes the refinement work is that Γ is independent of k. This is the identity $\sum_{l\geq 0} 2^l 3^{-l-1} = 1$: the l-th derivative contributes 2^l while the pole at $z = 3$ contributes 3^{-l-1}, and the two balance exactly.

The $\widehat{\gamma}_i$ are evaluated by an interval sweep in Arb using the closed form $G_2^{(2,0)} = G_2\,(z^2c^2 + zc')$, where $c = c(s,z)$ is as in §6.1; that closed form is proved rather than assumed. At the pinned settings — primes to $P = 10^5$, 800 boxes, 300 bits — the sweep gives $\Gamma = 104.5196$, as a head contribution 203.118 and a tail 5.921 against $\sup|G_2^{(2,0)}| \leq 19881.2$ on $|z| = 2.9$.

The refined estimate is what lowers the sweep threshold to $K_0 = 78$, and so what lets `sweep_mid` begin at $k = 79$ rather than higher; the unrefined first-order route of §6.1 would leave a gap between the sweeps.

Remark 6.2. Two routes through §6.3 are available, and we have taken the weaker one. The uniform bound (21) is self-contained and proved; it costs a larger analytic threshold, which the

certified sweeps then absorb. The δ-table route evaluates (20) and exploits the sign cancellation, giving a much smaller threshold at the price of a substantial certified computation. Since the sweeps cover the intervening range either way, the theorem does not depend on the choice, and we prefer the route that rests on less.

From `NumericalFacts`, with the values from the current certified run:

7. Assembly and the certified ranges

The proof of Theorem 1.1 splits $[6, \infty)$ into four ranges. The first three are certified computations; the fourth is the analytic argument of Proposition 5.1.

range	source	field
$6 \le k \le 199$	the prior certified f_k computation	`sweep_low`
$200 \le k \le 899$	refined criterion sweep ($K_0 = 78$)	`sweep_mid`
$900 \le k \le 2439$	extended δ-table sweep	`sweep_high`
$k \ge 2440$	derivative domination, proved here	`phi_at_k1`, `phi_mono`

The ranges overlap pairwise, so their union is $[6, \infty)$ and Theorem 1.1 follows.

For the analytic range, Proposition 5.1 reduces matters to positivity of φ. The certified scan gives $\varphi(2440) = 89 \pm 0.35 > 0$, and $\varphi(m) \le \varphi(m+1)$ for every $m \ge 2440$; induction gives $\varphi(k) > 0$, hence $f_k < f_{k+1}$, for all $k \ge 2440$.

Remark 7.1. The monotonicity of φ above 2440 is a statement about (16): the leading term $a_1 k^2$ grows quadratically while $R(k)$ grows linearly in k apart from the δ-table contribution, which is controlled by the same certified table.

8. A general principle

Conjecture 8.1. Let $\mathcal{P}$ be a set of primes with least element q, and let $f_k^{\mathcal{P}} = \sum 1/(n \log n)$ over those n with $\Omega(n) = k$ and all prime factors in $\mathcal{P}$. Then the k^2 term is present — and hence the sequence is eventually increasing — if and only if $q = 2$.

The heuristic is Proposition 3.1. The regularised product $H^{\mathcal{P}}(s, \cdot)$ is analytic in $|z| < q^s$, and the evaluation point relevant to the 2^{-k} scale is $z = 2$; so $z = 2$ lies on the boundary at $s = 1$ precisely when $q = 2$, and strictly inside otherwise. On the boundary the k^2 factor appears and competes with 2^{-k}; strictly inside it does not.

Theorem 1.1 is the case $\mathcal{P} = \{\text{all primes}\}$, $q = 2$, and Theorem 1.2 is the case $\mathcal{P} = \{p \ge 3\}$, $q = 3$. For $q \ge 3$ the argument of §4 applies with only the constants changing: the limit is the omitted Euler factors evaluated at $z = s = 1$, the coefficient of 2^{-k} is $H^{\mathcal{P}}(1,2)\,(\partial_s \log H^{\mathcal{P}}(1,2) + 2\gamma)$, and the remainder is controlled by Cauchy's estimate after factoring out the pole at $z = q^s$.

Two caveats keep this a conjecture rather than a theorem. The first concerns the "if" direction for general $\mathcal{P}$: the argument of §4 needs the regularised product to converge and its coefficients to be extractable, and while $\sum_{p \in \mathcal{P}} p^{-2s}$ converges for any $\mathcal{P}$, the identification of the limit and of the coefficient of 2^{-k} uses the full Euler product for ζ. For $\mathcal{P}$ the set of all primes above a bound, or a union of arithmetic progressions of primes, the missing factors are controlled and the argument should go through unchanged; for arbitrary $\mathcal{P}$ we do not claim it.

The second concerns the "only if" direction. That $2 \in \mathcal{P}$ places $z = 2$ on the boundary is immediate, but that the k^2 term is then genuinely present — rather than vanishing by some cancellation — is a statement we have verified only for $\mathcal{P}$ the set of all primes, where it is [2].

Remark 8.2. The two theorems of this paper are the two extreme cases: removing the single prime 2 already moves the radius past the evaluation point, and no smaller change does. In that sense the conjecture asserts that nothing between them occurs.

9. What is proved and what is assumed

We separate three things: what is proved in Lean, what is computed in ball arithmetic and taken as a hypothesis, and what is inherited from [2].

- The analytic reductions of §§3–4 and §§5–7 are formalised in Lean 4 with no `sorry`, resting only on `propext`, `Classical.choice` and `Quot.sound`.
- The numerical inputs are ball-arithmetic computations, taken as hypotheses rather than proved within the formalisation. For Theorem 1.1 these are collected in a structure `NumericalFacts`: the enclosures of d and ρ, the bounds $S^* \le 6.9763$, $\alpha_+ \le 15.4257$ and $\Gamma \le 104.52$, the aggregate source bound and the two φ facts at the analytic threshold, and the three sweeps $[6, 199]$, $[79, 899]$, $[900, 2439]$. For Theorem 1.2 the corresponding structure carries the enclosure of S, the bound on M_R, the expansion (10) with its remainder bound, and the sweep at small k.
- Consequently both theorems are proved *conditionally on* those computations, in the same sense and to the same extent. What the formalisation establishes is that no gap remains in the reduction: given the certified inputs, the analytic argument is complete and machine-checked.
- A second form of the main theorem replaces the aggregate source bound by the quantities the certified package actually computes — the β_i sweep, M_0, the S^* and α_+ partial-sum bounds, the U/V and $E_{\exp}$ sources — and derives the source bound from them, so that §§6.1–6.4 are load-bearing rather than merely stated.
- Two facts absent from Mathlib were proved in the course of the odd formalisation and may be of independent use: $\lim_{s\to 1^+}(\zeta(s) - 1/(s-1)) = \gamma$, and that $\zeta(s) - 1/(s-1)$ extends analytically across $s = 1$.

Appendix A. Certified computations

All interval computations use Arb/FLINT through python-flint. The Stieltjes constants entering the representation of $\zeta(s)(s-1)$ in §6.1, and the ζ evaluations used in boundary checks, are FLINT's own certified implementations. The δ-table of §6.3 was cross-validated against an independently written high-precision implementation of the original integral definition of A_m, at 40–80 decimal digits, agreeing to six or more significant figures at each test point.

The scripts are `numerics/` for Theorem 1.1 and `odd-proof/numerics/` for Theorem 1.2. The two sweeps behind α_+ and Γ use 1000 and 600 boxes respectively in $s \in [1, 2]$.

For the odd case, M_R is computed over 40 boxes in $s \in [1, 2]$ and 100 boxes in θ on the circle, with primes to 5×10^4 and an explicit tail bound. Subdividing in s is essential rather than cosmetic: a single box over $[1, 2]$ gives only $M_R \le 739605$.

The five constants of (12) beyond S are certified as follows.

constant	bound	method
S_0	≤ 44.5987	first-order pole; recursion (head), Cauchy (tail)
S_1	≤ 486.1945	second-order pole; recursion + Cauchy tail
C_N	≤ 367.8435	Moebius prime zeta, recursion + Cauchy tail
N_U	≤ 26.0962	exact head to $k = 400$, explicit envelope beyond
C_E	≤ 5.48910	Taylor model in $\varepsilon = s - 1$

All five are `arb` enclosures end to end, with no floating-point step in the chain, and two of them needed a change of method to get there. C_E is certified over the whole range $s \in (1, s(3)]$ rather than at a sample of points. The four constants making it up are ratios in which the pole of ζ at $s = 1$ cancels, so interval evaluation on boxes of s loses the cancellation and diverges however narrow the boxes are taken; expanding ζ as a power series about $1 + E/2$ makes the cancellation symbolic, and the resulting series — whose coefficients decay — evaluates tightly on the whole disc in one step. C_N requires the prime sums $\sum_{p\geq 3} p^{-t}$ for $t = s, 2s, \ldots, 200s$; enumerating primes to a cutoff will not do, since at $s \approx 1.05$ the convergence is slow enough that a cutoff of 10^8 with Rosser–Schoenfeld still leaves a factor of two, which compounds through the recurrence. The Möbius series $P(t) = \sum_n \mu(n) n^{-1} \log \zeta(nt)$ avoids primes entirely. The scripts are `s0_recursion.py`, `s1_recursion.py`, `s1_cauchy.py`, `cn_certified.py`, `nearu_certified.py` and `ce_taylor.py`.

The two sweeps behind α_+ and Γ are run at pinned settings: primes to $P = 10^5$ and 300 bits of precision in both, with 1600 boxes in s for α_+ and 800 for Γ. The f_k certificate records its own parameters — 2194 panels of width 0.8, 52 Gauss–Legendre points per panel, 1400 bits, Bernstein-ellipse parameter 1.45 — together with a SHA-256 hash of the payload excluding the hash field itself.

The quadrature error is controlled by [4, Thm. 19.3]: if f is analytic on $[-1, 1]$ and continues analytically to the open Bernstein ellipse E_ρ — the ellipse with foci ± 1 whose semimajor and semiminor axes sum to ρ — with $|f| \leq M$ there, then $(n+1)$-point Gauss quadrature satisfies, for $n \geq 2$,

$$\left| \int_{-1}^{1} f - \widehat{I} \right| \;\leq\; \frac{144}{35} \, \frac{M \rho^{-2n}}{\rho^2 - 1}. \tag{22}$$

The panel decomposition applies (22) on each of the 2194 panels, with ρ determined by the distance from the panel to the nearest singularity of the integrand.

The certified run records python-flint 0.9.0 under Python 3.12.3.

In the implementation the per-panel bound is applied in the equivalent form

$$10 \cdot \frac{64}{15} \, \frac{\rho^{-2N}}{1 - \rho^{-2}}, \qquad N = 52,$$

which coincides identically with (22) at $n = N - 1 = 51$, since $\rho^{-2N}/(1-\rho^{-2}) = \rho^{-2(N-1)}/(\rho^2 - 1)$; the constant $64/15$ is used in place of $144/35$, which is larger by 3.7% and hence conservative. With panel width $H = 0.8$ and ellipse parameter $\delta = 1.45$ one has $\rho = \delta' + \sqrt{1 + \delta'^2}$ for $\delta' = 2\delta/H$, giving $\rho \approx 7.385$ and a per-panel discretisation error below 10^{-88} — far smaller than the arithmetic and truncation errors it accompanies.

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